



Post-Quantum Cryptography Migration

State of Affairs, Part 1 of 2: Standards, Certificates, Networks and Policy

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

This report is part one of two on the state of the post-quantum cryptography migration. It covers the month from the last full edition, at the start of September, to 4 October. This part covers the standards, the certificates, the network protocols and the government policy that sit around them. Part two covers the libraries, the hardware, the platforms and the industry sectors.

The report opens with the handful of developments that change the picture. It then walks through the standards work at the IETF, followed by certificates and identity, where the biggest story of the month sits. Network protocols come next, from VPN tunnels to DNS, and government policy follows, from the G7 to the American federal identity card.

The report closes with a short account of where things stood before September. Readers who skipped the earlier editions can start there and come back.

1. What Matters Most

A month of updates holds a few developments that change the picture and many that only refresh it, and this section names the ones that change it.

Five things moved this month, and each one is a turn in its own story. A large network operator announced it will become a public certificate authority built for post-quantum certificates. The two core TLS drafts for ML-KEM and ML-DSA were frozen by appeals.

The composite certificate drafts kept edging toward publication. Group messaging reached a final review. And governments issued a fresh round of calls, deadlines and pilot programmes.



Start with the certificates, because that is where the migration's hardest problem lives. On 29 September Cloudflare announced it intends to become a publicly trusted certificate authority by acquiring root key material from GlobalSign [1]. It has applied to the Chrome, Apple, Microsoft and Mozilla root programmes [1]. The new authority will issue both classical certificates and Merkle Tree Certificates, a design that replaces a per-certificate signature with a short proof that the certificate sits in a signed log [1]. Cloudflare says standard Merkle Tree Certificate issuance will be free, and it targets Chrome's quantum-resistant root store by early 2027 [1].

Apple moved in the same week. On 21 September its root programme published its first post-quantum policy position [2]. It plans a Merkle Tree Certificate policy for web server authentication, with a draft expected by the end of October [2]. It will also accept composite-signed root certificates for secure email [2]. Apple's position joins Chrome's earlier stated preference for the same certificate design [3].

From certificates to the standards behind them. On 24 September the IETF's security area director recorded that complaints and appeals had been received about how the TLS working group handled its ML-KEM key exchange draft [4]. The document is held in the RFC Editor's queue while they are reviewed, with its status changed to blocked under a stream hold [4]. The companion draft for ML-DSA signatures in TLS was blocked the same day under the same hold [5].

Both core post-quantum TLS documents had cleared approval, and both are now frozen on process grounds. Nothing in the record says how long the review will take.

The composite constructions moved the other way. The composite ML-DSA signatures draft for certificates reached the second editing stage at the RFC Editor on 2 October [6]. Its sibling for composite ML-KEM remains in IESG evaluation with two DISCUSS positions, the formal objections that must be cleared before approval [7].

Messaging and the governments complete the list. The MLS group messaging working group opened last call on its post-quantum cipher suites on 1 October, running to 15 October [8]. And on 3 September France's cybersecurity agency announced that the G7's cybersecurity working group is issuing an urgent call to action on the transition [9].

2. Standards in Motion: TLS, Composites and Group Messaging

Standards are the agreements that let one vendor's software talk to another's, and a post-quantum algorithm is only useful once a protocol says exactly how to carry it.

Recent Developments

The month's standards news is a story of documents near the finish line, and two that stopped short of it. The TLS holds described above are the sharp exception. The RFC Editor state of the ML-KEM draft changed from in progress to blocked on 24 September [4]. The ML-DSA draft, version 6, moved to blocked the same day [5]. Its IANA registrations were acknowledged on 28 September while it stayed blocked [5], so the paperwork continues even as publication waits.

The hybrid key exchange that most of the web already uses is a separate document, RFC 10024, and it was published in August [10]. That RFC marks the X25519 hybrid group as recommended in the official registry, while the two other hybrid groups are not [10].

The composite certificate work is the furthest along. Composite signatures pair a post-quantum signature with a classical one, so the certificate stays secure if either algorithm holds. The composite ML-DSA draft for X.509 sits in the RFC Editor queue as a Proposed Standard [6]. It moved to a second editor on 23 September and into its second edit on 2 October [6]. A companion draft carrying the same composite signatures inside CMS, the message format beneath signed email, reached the same queue and moved to a second editor on 1 October [11].

Composite ML-KEM is one step behind. Its draft reached version 21 in September and entered the IESG state “Revised I-D Needed” on 3 September [12]. One DISCUSS was entered by an IESG member on 27 August against the previous version [12]. A second IESG member revised his own on 1 October, asking that the draft’s ASN.1 module cite a normative reference for its formal syntax [12]. That objection concerns a citation, not the cryptography.

Current State

The web tokens and signed objects that applications pass around are getting their own post-quantum forms. The JOSE and COSE draft for composite signatures reached version 4 on 11 September [13]. The COSE draft for SLH-DSA, the hash-based signature, went to the IESG in August and entered area director evaluation [14]. A new OpenPGP draft extends RFC 9980 with composite schemes on NIST and Brainpool curves [15].

Group messaging is close to its own standard. MLS is the IETF protocol for end-to-end encrypted group chat.

The MLS post-quantum cipher suites draft entered working-group last call on 1 October [16], with the call ending on 15 October [8]. The document under review is version 6, posted in July, aimed at Proposed Standard [17]. An earlier last call in May ended with a revision requested [16]. A companion draft that amortizes the cost of post-quantum MLS was refreshed in August [18].

Certificate enrollment moved a notch. The ACME profiles draft, which lets a certificate authority offer a menu of certificate profiles to clients, went to the IESG on 21 September [19]. The shepherd describes the working-group consensus as weak, with few respondents, all strongly in favour [19]. An individual draft would let certificate authorities promise continuity of post-quantum issuance across their own key rotations [20].

3. Certificates, Identity and Trust Stores

Certificates bind a public key to a name, and everything from web browsing to signing in at work depends on them, so this is the migration’s long pole.

Recent Developments

Cloudflare’s plan has more moving parts than the headline. The acquisition of GlobalSign root key material is expected to close within two months [1]. Issuance will run on a fork of Boulder, the open-source software Let’s Encrypt uses [1]. Cloudflare will run mirroring cosigners and launch a new family of certificate

transparency logs [1]. Its earlier Merkle Tree Certificate experiment with Chrome, which used classical signatures, wound down in August after serving billions of certificates [1].

Apple's expected policy is demanding. Each certificate would carry three cosignatures, including ML-DSA from the issuing operator and ML-DSA from a separate mirror operator [21]. Cosigner keys must sit in hardware validated to FIPS 140-3 Level 3 or Common Criteria EAL 4+ [21]. Certificates would be capped at seven days, with annual audits and published domain validation records [21]. Applications from operators are expected from late summer 2027 [21].

The certificate format itself kept evolving. The IETF Merkle Tree Certificates draft reached version 6 on 21 September [22]. It adds tree pruning and a fast issuance mode that uses witness signatures rather than waiting for the next checkpoint [22].

Current State

On the classical-certificate side, the vendors disagree about hybrids. DigiCert now recommends pure ML-DSA certificates as the default over composite ones [23]. It cites broad implementation maturity in OpenSSL, BoringSSL and Java, and the CA/Browser Forum's focus on ML-DSA [23].

The enterprise platforms are building both. EJBCA Community 9.6, released 16 September, issues composite ML-DSA certificates pairing a classical key with an ML-DSA key [24]. SignServer Community 7.7.1 adds composite certificate support too, though HSM key storage now requires its enterprise edition [25].

Other signing services filled in. HashiCorp Vault Enterprise 2.1.0 added SLH-DSA to hybrid sign and verify operations in its transit engine [26]. BeatQuantum launched a timestamp authority that signs with pure ML-DSA-87 or SLH-DSA, with no classical component [27]. wolfSSL can now sign certificate revocation lists with post-quantum keys [28]. Microsoft's certificate services documentation now extends its phased plan to the enrollment and status role services [29].

Sign-in is where the migration trails. Keycloak 26.8.0 shipped on 1 October without the open pull request that would add ML-DSA token signing [30]. A new issue asks Keycloak to handle signing algorithms outside its fixed list, which overlaps the ML-DSA work [31]. The passkey library fido2-net-lib, written for .NET, merged WebAuthn Level 3 with ML-DSA credential verification on .NET 10 [32].

The library's next preview was a security release that every preview user is told to install [33]. Giesecke+Devrient is adding prototype post-quantum protocols to its Java Card system for European identity cards [34].

Kerberos, the protocol behind most corporate logins, has a live draft. Version 2 of the post-quantum PKINIT draft adds ML-KEM key establishment and covers composite ML-KEM as well [35]. A client that signed with a post-quantum certificate must reject a server response signed with RSA or ECDSA [35]. That rule closes the obvious downgrade.

4. Networks: VPNs, SSH, DNS and the Web's Transport

Network protocols carry the data that can be recorded today and decrypted later, which is why key exchange moved first and why size is now the main engineering problem.

Recent Developments

The IPsec world spent the month on size. IKEv2, the protocol that sets up VPN tunnels, limits each payload to 64 kilobytes by a two-byte length field [36]. A draft that lifts that limit was adopted by the ipsecme working group on 29 September [37]. Its motivating example is Classic McEliece, whose public keys exceed 100 kilobytes [36]. The large payloads must stay out of the first exchange and ride in the intermediate exchange instead [36].

A second draft takes the transport route. Version 8, posted 1 October, runs IKEv2 over TCP while the tunnels it creates stay on ordinary unreliable transport [38]. That lets large post-quantum key material move without the performance cost of tunnelling over TCP [38]. Meanwhile the draft adding ML-DSA and SLH-DSA authentication to IKEv2 was approved by the IESG in August and entered the RFC Editor queue [39].

Implementations kept pace. Cisco's IOS XE 26.2 adds ML-DSA signatures for IKEv2 authentication, completing a fully post-quantum tunnel when paired with the ML-KEM hybrid [40]. strongSwan 6.1.0 hardened its ML-KEM code against empty shared secrets and disabled the old IKEv1 by default [41]. Libreswan 5.4, released in August, added ML-KEM-768 and the multiple key exchange framework [42].

SSH finished a standard and started on signatures. RFC 10042, published 31 August, defines the hybrid ML-KEM key exchange methods for SSH [43]. Go's SSH package added ML-DSA host keys and user keys on 29 September [44]. They sit last in the preference list, so adding one does not change the host key existing clients already know [44]. Go also added ML-DSA certificate algorithms under its own names, because no specification defines one yet [45].

Current State

DNS is still the hardest packaging problem. Cloudflare's 1.1.1.1 resolver now validates DNSSEC signatures made with ML-DSA-44, at 2,420 bytes each, and plans full post-quantum DNSSEC by 2029 [46]. On the client side, systemd-resolved gained ML-DSA-44 verification [47]. A new draft defines SigTag, a signal that lets a resolver say it already knows a Merkle tree so the server can omit the full signature [48]. VeriSign filed patent disclosures against it the same day [49].

The DNS community is still weighing its options. Paul Hoffman's considerations draft lists the tradeoffs and says it is not meant to become an RFC [50]. One competing draft for hash-based DNSSEC signatures expired on 1 October with no new version [51].

On the web's transport, Cloudflare's Automatic Key Exchange probes origin servers and prefers the hybrid group on the first handshake when the origin supports it [52]. Cloudflare says that raised post-quantum origin connections to about 45 billion a day and cut handshake retries from about 52 percent to 3.7 percent [52]. Its quiche 0.30.0 now resolves to Boring 5, which turns post-quantum groups on by default [53]. The release notes warn that this can split the first handshake message across several QUIC packets [53].

The libraries beneath these protocols shipped fixes. OpenSSL disclosed a high-severity DTLS flaw on 29 September, fixed across its supported lines [54]. OpenSSL 3.5.9 also fixed five low-severity QUIC issues [55]. The 4.1.0 beta adds DTLS 1.3 [56]. wolfSSL 5.9.4 added post-quantum-only TLS 1.3 builds, with no RSA, elliptic curve or Diffie-Hellman at all [28].

Network access control is starting too. A revised draft for the TLS-based EAP methods says deployments needing long-term confidentiality must use TLS 1.3 with a post-quantum or hybrid group [57]. The IAB's workshop on post-quantum authentication meets in Prague on 11 and 12 October [58].

5. Governments, Mandates and Federal Identity

Governments turn technical advice into dates with consequences, and those dates now drive vendor roadmaps as much as the cryptography does.

Recent Developments

The G7 spoke first. On 3 September France's cybersecurity agency announced the G7 Cybersecurity Working Group's urgent call to action for public and private decision-makers [9]. The group, led by France under its 2026 presidency, brings together the G7 agencies with the European Commission and ENISA as guests [9]. The call builds on a 2025 statement negotiated under Canada's leadership [9].

Europe's machinery kept turning. The EU published the results of its consultation on the coordinated post-quantum roadmap on 2 September [59]. The EU Quantum Act, once expected in the second quarter, is now tentatively planned for the fourth [60].

ETSI published a specification for authenticated quantum-safe hybrid key establishment [61]. Britain's consultancy pilot drew enough interest that expressions of interest closed, and it runs until March 2027 [62]. Germany's BSI advised against Classic McEliece in new developments, citing progress in cryptanalysis but no practical attack on the recommended parameters [63].

In the United States, the executive order sets the clock. Executive Order 14412 and memo M-26-15 direct civilian agencies to move key establishment on high-value systems to post-quantum algorithms by 31 December 2030 [64]. Digital signatures follow by 31 December 2031 [64]. The first phase, through 2027, covers strategy, planning and inventory [64]. On the national-security side, NSA says all new commercial systems must support CNSA 2.0 by 2027, with legacy systems phased out by 2030 [65].

Certification schemes attached their own dates. NIAP's Policy Letter 33 says products failing CNSA 1.0 for every cryptographic function will not be accepted into evaluation from 1 January 2027 [66]. From 1 July 2027 they will not be listed [66].

Chainguard's FIPS provider became a validated FIPS 140-3 module with hybrid post-quantum key exchange as an approved service [67]. Red Hat advises not to expect the FIPS and post-quantum situation to stabilize before 2028 [68]. The new PCI key management standard does not mention quantum computing at all [69].

Current State

Federal identity is where policy meets a card in a wallet. GSA is updating its identity architecture, FICAM, for post-quantum algorithms with an emphasis on crypto agility [70]. The interagency working group the OMB memo required met first on 12 August, with 40 people from 17 agencies [70]. Trade coverage gives the phases as planning through 2027, pilots through 2028 and full migration by 2035 [71].

The testing laboratory is the gate. GSA is expanding its FIPS 201 laboratory to test quantum-resistant badges, visitor passes and access controls [70]. Agencies may only buy access control equipment on its approved list, which makes that testing a procurement requirement [72]. As of 11 September the list held no product explicitly supporting ML-KEM or ML-DSA [73].

The PIV standards themselves move slowly and in public. NIST released working drafts for post-quantum PIV in June, with no comment deadline [74]. Its drafting repository has had nothing pushed since 21 May [75]. Its open issues come from outside contributors, including whether on-card ML-DSA signing should be deterministic or hedged [76].

A new issue reports that a table of algorithm identifiers is missing from the SP 800-78 draft [77], and a fix was still unmerged on 4 October [78]. NIST also contracted Electrosoft in August to advance post-quantum work in federal identity [79].

GSA's summit on 16 September carried that agenda. It was open only to government personnel, and all nine named speakers held government posts [80].

Unresolved Conflicts

Two disagreements sit in this section. NIAP's policy letter makes CNSA 2.0 mandatory for evaluations from 1 January 2028 [66]. NSA's earlier CSfC addendum gives 2028 as the year CNSA 2.0 is required in capability packages, with all solutions by 2030 [81]. These may describe different gates, and the final addendum text would settle it.

The second is smaller. PQSecure says on its own events page that it presented hardware and software products at the GSA summit [82]. GSA's published programme names no industry speaker or exhibitor [80]. A recording or official attendee list would settle it.

6. Where Things Stood Before This Update

This section gives the earlier picture for readers who skipped the previous editions, so that the month's news has something to stand against.

Before September, the picture split cleanly in two. Key exchange had crossed from experiment to majority. Signatures and certificates had barely started.

On the network side, post-quantum key exchange rose from about 29 percent of one network's HTTPS requests in mid-2025 to 54 percent in mid-2026 [83]. RFC 10024 defined the three hybrid groups for TLS as a Proposed Standard in August [84]. Akamai reached general availability for end-to-end post-quantum connections across its network the same month [85]. OpenSSH had made post-quantum key agreement its

default years earlier, and since version 10.1 it warns users when a connection falls back to classical [86]. In IPsec, RFC 9370 already allowed several additional key exchanges in one tunnel setup [87].

Certificates were the opposite story. The count of ML-DSA roots in the Mozilla, Apple, Microsoft and Chrome stores was exactly zero [88]. The industry had chosen Merkle Tree Certificates as the way out, with Chrome naming them its preferred path and Cloudflare running a feasibility experiment [3].

Behind the public web, enterprise authorities were further along, and AWS's private CA had issued ML-DSA certificates since November 2025 [89]. The identity layer trailed everything, with no specification for post-quantum SAML signatures [90]. WebAuthn Level 3 became a W3C Recommendation in August without post-quantum algorithms [91].

Messaging led the whole migration. Signal had extended post-quantum protection into its ongoing message ratchet [92]. Apple's iMessage had run post-quantum rekeying since early 2024 [93]. Email had a new standard in RFC 9980, which brought post-quantum algorithms into OpenPGP in June [94]. GnuPG, the most widely used implementation, followed an incompatible specification instead [95].

Governments had converted guidance into dates. Memo M-26-15 set a five-phase federal timeline running from 2026 to 2035, with agency plans due within 120 days [64]. France's cybersecurity agency said it would stop certifying security products without quantum-resistant encryption from 2027 [96]. The EU roadmap asked member states to begin by the end of 2026 and to move critical infrastructure by the end of 2030 [97]. Britain set phases running to 2035 [98].

The regimes also diverged: the American suite prefers pure post-quantum, France mandates hybrids, and Germany recommends algorithms NIST did not choose [96]. No single implementation satisfied all three, and that was the state of play when September began.

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Post-Quantum Cryptography Migration

State of Affairs, Part 2 of 2: Libraries, Hardware, Platforms and Sectors

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About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. It uses several AI models from competing labs, running on different platforms. One model gathers the current claims on a topic, a different one checks each claim against its source, and disagreements are kept, not smoothed over. The aim is an even-handed account that no outside party has shaped. AI makes mistakes, and so does the system built around it. We find and fix some in every cycle, so the methods keep changing and the content should improve over time. Updates come at least quarterly, sooner if something major happens.

This report is part two of two. Part one covered standards, certificates, networks and policy. This part covers the rest of the post-quantum migration: the libraries, the hardware beneath them, the platforms that ship them, and the sectors that depend on them. It runs from the last full edition, in early September, to 4 October.

It opens with the handful of developments that change the picture. From there it moves through the cryptographic libraries, then down into silicon, firmware and key storage. It climbs back up through operating systems, runtimes and clusters, then out to cloud services and network appliances. Discovery tools and the sectors at the edge come next.

It closes with a short account of where things stood before September, for anyone who skipped the earlier editions.

1. What Matters Most

A month of releases holds a few changes to the shape of the migration and many that only refresh it, and this section names the ones that change it.

The short version is that post-quantum cryptography became the default in more places and the only option in a few. One major library can now build TLS with no classical public-key algorithm at all. Windows can use ML-KEM on its own, without a classical partner. Kubernetes can stand up a whole test cluster on post-quantum certificates. The largest edge network says most browser traffic it sees is already protected.



Start with the libraries. wolfSSL 5.9.4, released in late September, can build TLS 1.3 using ML-KEM for key exchange and ML-DSA or SLH-DSA for authentication, with no RSA, elliptic curves or Diffie-Hellman [1]. It also added native Falcon and FrodoKEM [1]. OpenSSL 4.1.0 reached its first beta on 23 September with DTLS 1.3 and faster code for all three NIST algorithms [2]. Patch releases for the 4.0, 3.6 and 3.5 lines followed on 29 September [3].

On the desktop, Windows moved past hybrid-only key exchange. A September preview update says ML-KEM can now be used as a standalone algorithm for TLS key exchange, alongside the hybrid groups added earlier [4]. Hybrid groups pair a classical algorithm with a post-quantum one, so a standalone option rests on the new mathematics alone. An August preview for Windows 11 had already brought the same option to Schannel, the Windows TLS engine [5].

In the data centre, Kubernetes merged an option on 1 October that runs a local cluster entirely on ML-DSA certificates [6]. With it set, every TLS endpoint, from the API server to the kubelets, uses ML-DSA-65 keys by default [6]. The last edition found no source describing kubelet certificates with a post-quantum key at all. HashiCorp's Consul 2.1.0 release candidate now adds the hybrid group automatically whenever TLS 1.3 is the minimum [7].

At the edge, Cloudflare now shows customers which of their connections use hybrid ML-KEM. Its Radar data puts that at roughly 70 percent of browser traffic but only about 15 percent of origins [8]. The gap is the story. Browsers have moved, and the servers behind websites have mostly not.

Key storage passed a deadline of its own. 21 September was the sunset date for FIPS 140-2, the older US standard for validating cryptographic modules [9]. From here the validations that matter are under FIPS 140-3, and the post-quantum certificates are arriving there. Thales's Luna T7 holds a Level 3 certificate covering ML-KEM, ML-DSA and LMS, and the AWS-LC 3 module now holds issued certificates covering ML-KEM [9].

One security note belongs here too. liboqs, the Open Quantum Safe project's library, published an advisory on 22 September for an out-of-bounds read when checking LMS and HSS signatures [10]. The advisory names version 0.17.0 as the fix, but no such release exists yet [11]. The library discussion below gives the detail.

2. Libraries and Providers

Almost no application implements cryptography itself, so a few dozen libraries decide when software actually becomes quantum-resistant.

The libraries had a busy month, led by wolfSSL and OpenSSL. wolfSSL 5.9.4 brought SLH-DSA authentication to TLS 1.3 and DTLS 1.3 for all twelve parameter sets [1]. It replaced its Falcon wrapper around liboqs with a native implementation accelerated for Arm [12]. It added AVX-512 assembly for ML-KEM and ML-DSA [1], and a small-memory ML-DSA option proven on TI's C2000 signal processors [12]. The release also fixed eleven CVEs, three rated high, all in certificate and revocation handling and none in post-quantum code [12].

Hardening sits beside the features. The same release made an ML-DSA check constant time, wiped leftover ML-KEM and ML-DSA private key data, and added bounds checks to LMS and XMSS [12]. wolfSSL's own benchmarks on NXP's i.MX95 put its post-quantum code at 2.3 to 4.4 times faster than OpenSSL 3.5.7, a vendor measurement [13]. Its wolfCOSE library added LMS signatures for firmware and attestation use [14].

OpenSSL's 4.1.0 beta adds optimised ML-KEM and ML-DSA arithmetic for POWER, mainframe and x86-64 processors [15]. Amazon's fork, AWS-LC, shipped three releases. Version 5.9.0 added ML-KEM to HPKE, a standard way to encrypt a message to a public key [16]. Version 5.11.0 reads TLS groups from the system crypto policy and stops a crash on processors with AVX2 but no BMI2 [17] [18]. Google's BoringSSL sped up both algorithms [19], and Cloudflare Workers, built on it, added ML-KEM and ML-DSA to the Web Crypto API on 1 October [20].

The Java world caught up in one release. JDK 27 reached general availability on 15 September with hybrid key exchange for TLS 1.3, and X25519MLKEM768 is among its preferred groups by default [21]. Faster Curve25519 arithmetic lifts hybrid key exchange throughput by 27 to 51 percent in software [22]. New HotSpot intrinsics speed up ML-KEM, ML-DSA and LMS by up to 3.18 times over pure Java [23]. Bouncy Castle Java 1.86 closed timing leaks across MAYO, SNOVA and HQC [24], including a key-leaking flaw in its HQC code [25].

Mozilla's NSS 3.130 shipped on 23 September with a policy update for the ML-KEM-1024 group [26]. The Open Quantum Safe project spent the month on release candidates. oqs-provider, which plugs liboqs into OpenSSL, published 0.12.0 candidates on 8 and 16 September [27] [28]. The second adds the MQOM signature, removes SPHINCS+, and fixes a use-after-free in decapsulation [28]. No final 0.12.0 had appeared by 4 October [29].

A change merged on 1 October drops one hybrid from oqs-provider for OpenSSL 4.0 onwards, because OpenSSL itself plans to support it in 4.2.0 [30]. The liboqs advisory itself is moderate. A too-short LMS or HSS signature makes the checker read past its buffer [10]. The stray bytes feed only an internal hash, so the main risk is a crash [10]. Versions 0.11.0 to 0.16.0 are affected, and the fix was merged on 9 September [10] [31].

The plumbing between libraries and hardware filled in. pkcs11-provider 1.3.0, which connects OpenSSL to hardware tokens, added SLH-DSA and better ML-DSA handling on 22 September [32]. OpenSC's work to carry post-quantum keys in its smart-card driver, tested with a YubiKey and ML-DSA, was still unmerged on 4 October [33].

Unresolved Conflicts

Three disagreements sit in this section. The liboqs advisory names 0.17.0 as the patched version [10]. The releases page shows 0.16.0 as the latest, with the fix only on the main branch [11]. A 0.17.0 release, or an amended advisory, would settle it.

A SUSE advisory says liboqs 0.16.0 updated HQC to the August 2025 specification and turned it on by default [34]. The last edition held that HQC stayed off by default pending that specification. The liboqs 0.16.0 release notes would settle it.

Bouncy Castle 1.86 is described two ways. One page dates it 11 September and calls it a hardening release [24]. The release announcement dates it 15 September and lists new features, among them ML-KEM in CMS and S/MIME, HPKE with X-Wing, and LMS and XMSS in core [25]. Both may describe the same release, and the project's own changelog date would settle it.

3. Silicon, Firmware and Key Storage

Hardware decides how fast the new algorithms run and what a machine trusts before its software starts, and it is the hardest layer to change after it ships.

Processor support moved on several fronts. Intel's multi-buffer crypto library, intel-ipsec-mb, reached 3.0.0 on 1 October with ML-KEM and ML-DSA for all standard parameter sets [35]. The last edition had it accelerating only classical algorithms. The release adds SHA-3 and SHAKE with AVX-512 code, and a test provider that lets OpenSSL call it [35]. AWS-LC imported AVX2 assembly for ML-DSA from the mldsa-native project [36].

Open instruction sets moved in parallel. The RISC-V post-quantum extension ran its internal review from 11 to 25 September [37]. It targets the Keccak hash at the heart of these algorithms, and arrived with simulator and QEMU patches [38].

Secure elements and chip designs followed. Samsung is sampling the S3SSE2A, which it calls the industry's first secure element with ML-DSA in hardware, and puts the speedup at 17 to 18 times over software [39]. Infineon's PSOC Control C3 microcontrollers target CNSA 2.0 firmware protection, with samples available [40]. Rambus announced a root of trust supporting Caliptra, the open root-of-trust design for data centre chips [41].

Caliptra's code moved fastest. The 2.1 specification adds ML-KEM and an external-mu ML-DSA mode to its Adams Bridge accelerator [42]. SPDm is the protocol a server uses to check the firmware of the components inside it. Caliptra's microcontroller software now signs SPDm measurements with ML-DSA and issues ML-DSA-87 attestation tokens [43] [44]. It also prefers ML-KEM-1024 for SPDm key exchange [45].

OpenTitan, the open-source silicon root of trust, documents masked and fault-hardened ML-DSA and ML-KEM in its Earl Grey 2 datasheet [46]. A change merged on 12 September endorses its device identity certificates with ML-DSA during provisioning [47]. DMTF fixed a classical certificate-chain flaw in libspdm, the reference SPDm library, in releases on 29 September [48] [49].

Boot and update signing kept pace. wolfBoot 2.10.0 adds NXP's i.MX RT700 with an ML-DSA-87 signed-boot configuration [50]. wolfTPM 4.2.0 adds TPM-backed ML-DSA authentication for post-quantum TLS, plus support for SEALSQ's QVault TPM [51]. A classical deadline sits just ahead as well. Microsoft's certificate that signs the Windows boot loader expires on 19 October, replaced by its 2023 UEFI authority [52].

Key storage turned on validations and services. The AWS-LC 3 certificates replace that module's earlier in-process status [9], and AWS-LC 4 has joined the in-process list [53]. Go's cryptographic module also holds a Level 1 certificate covering ML-KEM [9]. AWS KMS now keeps ML-DSA keys inside FIPS 140-3 Level 3 hardware, where the last edition had only a CloudHSM preview [54].

Quantum key distribution stayed a separate market, with field trials rather than migrations. India sent keys over a 5.56 kilometre open-air link between two research institutions [55].

4. Operating Systems, Runtimes and Clusters

When a platform ships an algorithm, every application on it inherits it, which makes operating systems and runtimes the quiet engine of the migration.

Windows led the operating system news. Standalone ML-KEM key exchange came first to Windows 11 in an August preview (KB5120998) [5]. A September preview for OS build 28000 states the same capability (KB5124006) [4]. SymCrypt, the library beneath Windows cryptography, published version 103.13.0 on 4 September [56]. Its changelog lists HPKE for that version, after composite ML-KEM and composite ML-DSA in the two before it [57].

Apple and Google moved through their developer kits. Apple's open-source Swift Crypto 5.0 aligns with the CryptoKit in Xcode 27 [58]. It adds one-time ML-KEM and X-Wing private keys that can decapsulate only once, which stops a key being reused by mistake [58]. ChromeOS 152 began moving component and extension signatures from ECDSA to post-quantum algorithms [59]. Conscrypt 2.7.0, Android's TLS provider, made the hybrid group the default and added composite ML-DSA signatures [60].

Linux and the small systems filled in the rest. Mainline Linux carries ML-DSA in its kernel crypto library but no ML-KEM as of 4 October [61]. An open Zephyr module brings ML-KEM and ML-DSA to microcontrollers, with some post-quantum operations beating classical baselines in its benchmarks [62].

The runtimes and servers above them filled in defaults. Rustls 0.23.44 accepts ML-DSA certificates by default with the AWS-LC provider [63]. F5's NGINX-based delivery service on Google Cloud runs hybrid key exchange by default [64]. Redis merged a setting for TLS groups in August, and an open change documents the post-quantum setup [65] [66].

That Redis change carries two useful measurements. The hybrid group grows the ClientHello from 283 to 1,461 bytes [66]. OpenSSL clients send a key share only for the first listed group, so two hybrid-capable peers can quietly settle on classical if classical is listed first [66]. Apache Kafka has an open change adding an ordered list of key exchange groups, hybrids included [67].

Clusters and meshes moved from inheritance to intent. The Kubernetes option lets a different ML-DSA level be chosen [6]. Consul's release candidate adds explicit curve settings for sidecar mutual TLS and for agent traffic [7]. The last edition had only an open feature request.

Software signing gained its first post-quantum pieces in Go. go-tuf, the Go implementation of The Update Framework, merged ML-DSA support on 14 September [68]. It requires Go 1.27 and builds on Sigstore's signer code [68]. The Sigstore Go library 1.11.0 added native ML-DSA keys [69].

Unresolved Conflicts

Two disagreements sit in this section. Envoy's published documentation says its BoringSSL backend includes the hybrid group in the default curves, in both directions and in FIPS builds [70]. The source file on Envoy's main branch says the group is supported but must be opted in [71]. The release notes of the version that changed the default would settle it.

Red Hat's September roadmap lists ML-DSA support in Sigstore's cosign as an upcoming milestone [72]. The Sigstore library has already shipped ML-DSA keys [69]. A library and a command-line tool can differ, and a cosign release note would settle it.

5. Cloud, Edge and Network Appliances

A single cloud or appliance decision moves more traffic than a thousand enterprise projects, which makes this the fastest-moving layer of the migration.

AWS did its work in s2n-tls, the TLS library behind its service endpoints. Version 1.7.9 gave three older CloudFront policies a path to post-quantum, replacing empty KEM preferences with three hybrid ML-KEM groups [73] [74]. Version 1.7.11, released 2 October, adds 39 load balancer, KMS and service policies [75]. They include post-quantum and FIPS post-quantum families for Elastic Load Balancing [76].

Cloudflare had the busiest month. Its Automatic Key Exchange cut handshake retries to its origins from about 52 percent to 3.7 percent [77]. About 0.34 percent of origins fail when the larger key share leads, mostly behind middleboxes that drop multi-packet ClientHellos [77]. On 29 September it put IPsec downgrade protection into beta for Cloudflare WAN and Magic Transit [78]. Each peer signs the whole exchange, so an attacker cannot quietly force two post-quantum endpoints back to classical [78].

The other clouds shifted less visibly. IBM is retiring Hyper Protect Crypto Services, which had offered Dilithium signing [79]. New instances stopped in March 2026, remaining ones are deleted after 28 March 2027, and Key Protect is the named path [79]. Azure's Key Vault Premium and Managed HSM firmware moved to FIPS 140-3 Level 3 [80]. Google's load balancer release notes show no start to the second phase of its post-quantum rollout as of 4 October [81].

Appliances moved quickly, mostly in key exchange. OPNsense 26.7.4 updated strongSwan and now offers a short recommended list of post-quantum IPsec key exchanges [82]. Its August release offered a hybrid ML-DSA-44 and Ed25519 host key for SSH [83]. Cisco's Catalyst 9800 wireless controllers serve their web interface over TLS 1.3 with ML-KEM [84].

Wi-Fi took its first formal step. IEEE 802.11bt, the post-quantum Wi-Fi amendment, produced Draft 1.0 and entered its first working group ballot [85]. The Wireless Broadband Alliance sent the group a liaison on threats and use cases [86].

Unresolved Conflicts

Three disagreements sit in this section. Google's Cloud KMS release notes date the preview of quantum-safe key import to 6 August [87]. The last edition dated the announcement 20 August. Release notes and announcements often differ, and Google's own announcement date would settle which one is meant.

Netskope's earnings call describes post-quantum algorithms built into more than 120 data centres in the second quarter [88]. A product note describes commercial support behind a tenant-level feature switch [89], and earlier documentation called it a beta. Release documentation would settle it.

A Citrix best-practice paper says both default NetScaler SSL profiles bind the hybrid curve [90]. Citrix's product documentation says the feature works only in enhanced profiles [91]. The current release documentation would settle it.

6. Tooling, Sectors and Devices

Every migration starts with finding the cryptography and ends at devices and sectors with their own deadlines and hardware that cannot easily change.

Discovery tooling kept growing, though its biggest standard slipped. CycloneDX 1.7.2 fixed the XML schema so protocols can point to related cryptographic assets [92]. Its 2.0 milestone, due 31 August, was still open on 4 October with 92 open issues [93]. CBOMkit 2.3.0 added user-defined compliance policies [94].

Test tooling followed. testssl.sh can now detect the hybrid group when built on LibreSSL 4.3 [95]. The Open Quantum Safe test server moved to liboqs 0.16.0 and the oqs-provider 0.12.0 candidate branch [96]. Intel's 3.0.0 library added a build option that reports secret-dependent branches in its ML-KEM and ML-DSA code through Valgrind [35].

Satellite navigation produced the sector's clearest first. On 16 September, Galileo produced the first civil position authenticated through both navigation data and ranging signals [97]. Under controlled spoofing in Norway, the authenticated receiver held its position while conventional receivers were deceived [97]. The earlier plan put encryption of that signal across the whole constellation in the second half of 2026 [98].

Telecom moved from study to rules. 3GPP's security group began normative work on the first phase of post-quantum migration in Release 20 [99]. Its study report reached version 20.1.0 on 28 September [100]. In cars, a Thales paper found lattice schemes the best balance for vehicle-to-everything links and PKI-heavy traffic [101]. It also warns that legacy 32-bit car chips without crypto accelerators struggle [101].

Healthcare and industry stayed early. HHS's preparedness office is preparing post-quantum guidance for hospitals [102]. SEALSQ described secure chip design for implants and heart monitors, but named no medical customer and no specific post-quantum algorithm [103]. A Thales deck for Singapore's OT conference showed hybrid IPsec protecting remote access to a simulated controller, in a lab only [104]. It warns that controller makers cannot ship compliant post-quantum firmware until the industrial protocol standards move [104].

In finance, a payments study retired an old worry. SWIFT authentication runs out of band, so no post-quantum signature is too large for its messages [105]. That undoes an earlier recommendation of Falcon-512 to fit the message limit [105].

Unresolved Conflicts

One disagreement sits here. The last edition recorded a SEALSQ partnership with the Wi-SUN Alliance for smart-meter security. SEALSQ's half-year results name smart energy only as an application area, with no deployment or pilot [106]. A Wi-SUN or SEALSQ project announcement would settle it.

7. Where Things Stood Before This Update

Anyone new to this series needs the earlier picture, and this section gives it as background to everything above.

Before September, the migration had reached its defaults phase. OpenSSL 3.5 shipped the three NIST algorithms natively and preferred hybrid key exchange out of the box [107]. The attacks had moved from mathematics to code quality. One paper forged ML-DSA signatures from implementation bugs in under a second on one core [108].

In hardware, Crypto4A's module was described as the first quantum-safe HSM to pass FIPS 140-3 Level 3, in August [109]. The Trusted Computing Group's TPM 2.0 library version 185 was its first with ML-KEM and ML-DSA [110]. KMIP 3.0, the key manager interoperability standard, was nearing release with quantum-safe conformance levels [111].

Operating systems split on defaults. Apple enabled quantum-secure TLS for all system services across its 26-generation platforms [112]. Windows made hybrid groups available but left them disabled until administrators switch them on [113]. Edge, from version 147, made post-quantum key agreement mandatory [114]. Envoy, the proxy inside most service meshes, had closed its post-quantum issue as not planned [115].

Software signing had real firsts. Red Hat began dual-signing RHEL packages with a hybrid ML-DSA-87 and Ed448 key [116]. Android 17 introduced a hybrid APK signature pairing a classical key with ML-DSA [117]. Sigstore had tied its public services to ML-DSA landing in Go's standard library [118]. Linkerd made ML-KEM-768 the default for all meshed traffic [119].

The clouds were racing. AWS turned hybrid TLS on by default across KMS, Secrets Manager, ACM and S3 [120]. Azure's key services were still classical by the best available reading [121]. Akamai reached end-to-end post-quantum protection in August [122]. Middleboxes that assume a ClientHello fits in one packet were the known failure point [123], and Wi-Fi's standards work had only just begun [124].

The long tail was visible. LTO-10 became the first post-quantum-ready tape drive [125], and SAP set a readiness target of 2029 [126]. A post-quantum signature on the Eurosystem's TARGET2 platform came out thirteen times larger than RSA and overflowed its buffers [127]. One ML-KEM handshake could fragment into 62 LoRa packets [128].

Finding the cryptography had become policy. A June executive order made the cryptographic bill of materials a defined federal artifact [129]. One paper recovered a full ML-KEM key from fifteen power traces in under ten minutes [130]. The algorithms held, and unprotected implementations leaked anyway. That is the picture this update moves forward.

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